Remaining useful life, and how far ahead you asked

Two independent estimators on the NASA C-MAPSS turbofan run-to-failure fleets, with coverage reported as measured

On the engines closest to failure the sensor-driven estimator is 4.4 times more accurate than the age-only one, whose nominal 50% interval covers 11% of them.

FD001: 100 censored engines predicted from a fleet of 100 run to failure. Pooled mean absolute error 32 cycles for the age law against 20 for the degradation signal. On FD004, the hardest fleet, the age law's nominal 90% interval covers 53% of the engines nearest failure.

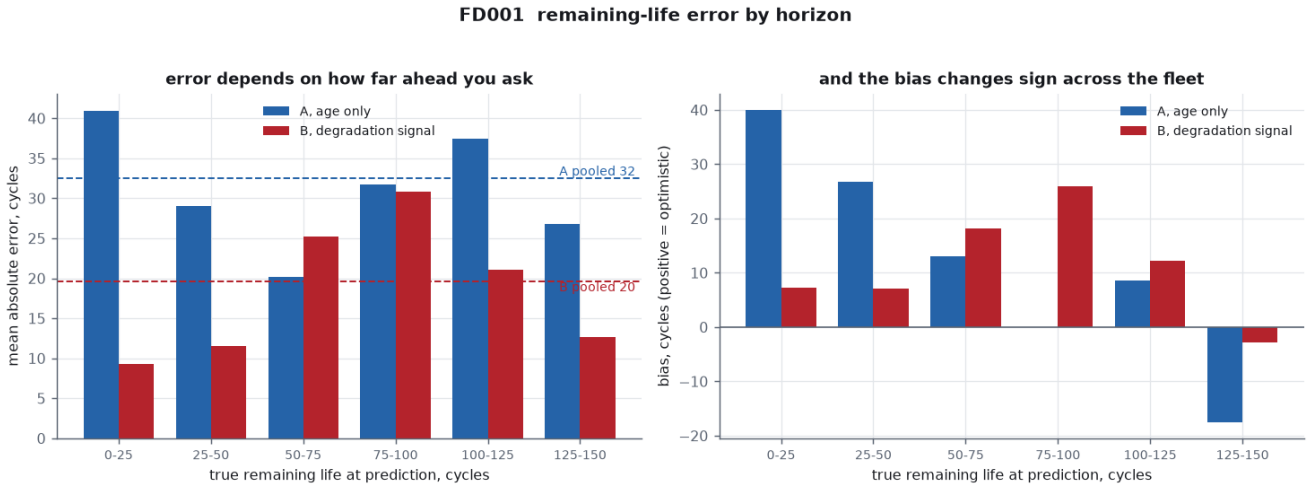
9 cycles
B near failure

41 cycles
A near failure

48%
A cover 50%, pooled

11%
A cover 50%, near failure

Remaining-life accuracy depends almost entirely on how far ahead the question is asked, so a single error number for a fleet is close to meaningless. Every figure here is reported against the true remaining life, and every prediction interval is reported at the coverage it was measured to achieve rather than the coverage it claims.



The question, and why one number cannot answer it

FD001, 100 censored engines

An engine is on the wing and the question is how much longer it can stay there. The C-MAPSS archive supplies four simulated fleets: one set of engines run to failure, and a second set whose records stop before failure, with the true remaining life supplied separately. So predictions can be scored against truth, and the censored half is genuinely censored rather than merely incomplete.

The difficulty of the question is not constant. Saying that an engine twelve cycles from failure has about twelve cycles left is a different task from saying the same of one two hundred cycles out, and the two errors point in opposite directions. Pooled over this fleet the age-only estimator's bias is +12 cycles, which is the average of +40 on the engines closest to failure and a negative bias on those furthest from it. It describes neither group.

true RUL	n	MAE	bias	optimistic	cover 50%	cover 90%
0-25	19	41.0	+40.0	95%	11%	100%
25-50	11	29.0	+26.7	91%	55%	100%
50-75	13	20.2	+13.0	69%	77%	100%
75-100	23	31.8	-0.0	48%	48%	96%
100-125	23	37.5	+8.5	65%	48%	96%
125-150	11	26.8	-17.6	27%	73%	91%
all	100	32.5	+12.3	66%	48%	97%

Error and measured coverage for the age-only estimator, banded by true remaining life. The last row is the pooled figure, printed to show what it conceals rather than instead of the bands.

Where the data came from

composition: what the files hold against what the readme claims

The dataset is NASA's Commercial Modular Aero-Propulsion System Simulation run-to-failure study. Its canonical home at the Prognostics Center of Excellence no longer serves the file: the URL redirects to a landing page with no download on it. The copy used here is the PHM Data Challenge S3 mirror, and the SHA-256 digests of both zip layers are pinned in the code, so a substituted mirror fails loudly instead of quietly changing the numbers.

Checking the parse against the archive's own readme rather than trusting it turned up an error in the documentation. The readme gives FD004 as 248 training and 249 test units; the files hold 249 and 248. RUL_FD004.txt settles it with 248 values, one per test unit, so the readme has the pair transposed.

subset	train found	test found	train in readme	test in readme	
FD001	100	100	100	100	agrees
FD002	260	259	260	259	agrees
FD003	100	100	100	100	agrees
FD004	249	248	248	249	readme is wrong

The operating-regime count is also recovered rather than assumed, and agrees with the readme's ONE and SIX. Within a regime the throttle is held exactly constant and the altitude and Mach settings carry about 0.003 and 0.0006 of dither, so the regimes are discrete flight conditions and not fuzzy clusters.

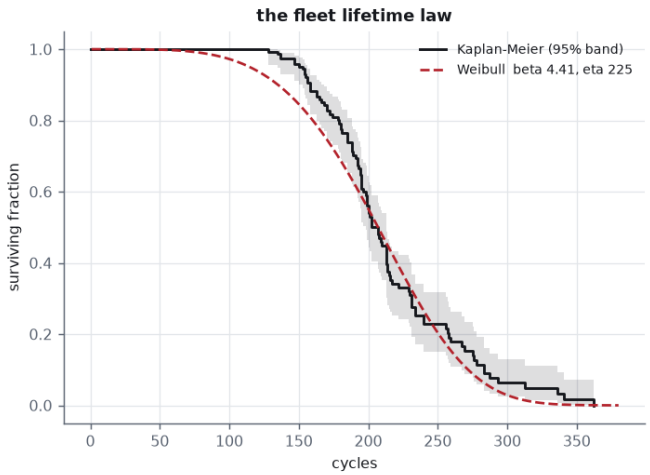
Estimator A: the fleet lifetime law

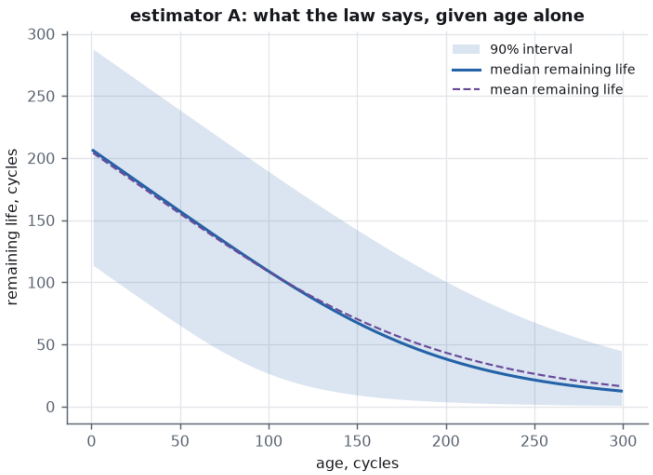
FD001, fitted on the 100 run-to-failure engines

The first estimator is told an engine's age and nothing else. A two-parameter Weibull is fitted to the fleet's lifetimes by right-censored maximum likelihood, and remaining life follows from conditioning that law on survival to the engine's current age. The inversion is closed-form, so the point prediction and the interval come from one expression with no simulation in it.

The fitted shape is 4.41, comfortably above one and with a confidence interval excluding it, so the fleet is in wear-out: hazard rises with age, which is what a growing fault should look like. The scale is 225 cycles.

This estimator is deliberately the weaker of the two and deliberately the one that cannot be accused of leaking anything. Two engines of the same age always receive the same answer, so its error has a floor set by the spread of the lifetime distribution itself. It is the floor the second estimator has to beat.





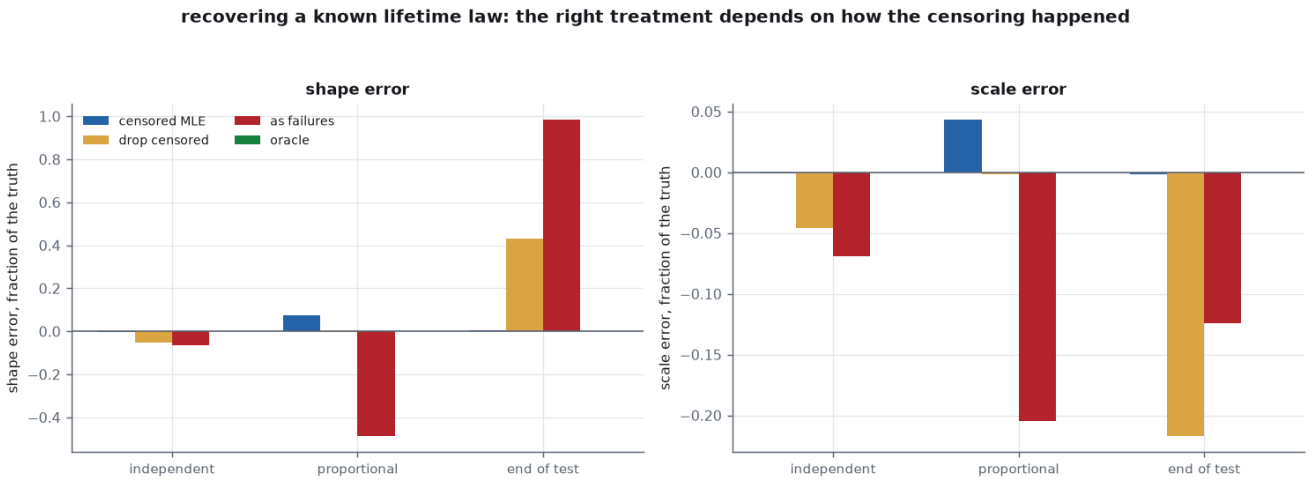
Censoring: the right treatment depends on the mechanism

recovering a known lifetime law, 40,000 simulated units

Half of each pooled fleet is right-censored. Those records are not missing data; each one says "this engine was still running at cycle 134", and the censored likelihood is how that gets used. Because C-MAPSS supplies the censored units' true failure cycles, the cost of mishandling them can be measured against an oracle rather than argued about.

The measurement did not agree with the textbook. Discarding the censored units, supposedly the blunder, came closest to the oracle, and the correct likelihood came out biased long by five to fifteen per cent on the scale. Simulation from a planted truth explains both. C-MAPSS cuts each test record at a random fraction of its own life, so the cut-off correlates with the true lifetime at +0.65 here — which is precisely what non-informative censoring rules out, and it biases the likelihood long. And because the censored units are a random half of a fleet whose lifetime distribution matches the half retained, discarding them costs precision and biases nothing.

> This is not a licence to drop censored data. Under the censoring an operator actually faces — a common cut-off, where the censored engines are the long-lived tail — the same shortcut recovers a shape 43% too high and a scale 22% too low, while the censored likelihood stays within half a per cent. Across all three schemes the censored MLE is the only treatment never badly wrong.



treatment	indep. shape	indep. scale	propn. shape	propn. scale	cut-off shape	cut-off scale
censored MLE	-0.1%	-0.1%	+7.7%	+4.3%	+0.3%	-0.2%
drop censored	-5.3%	-4.6%	-0.3%	-0.2%	+43.2%	-21.7%
as failures	-6.3%	-6.9%	-48.8%	-20.4%	+98.3%	-12.4%
oracle	-0.1%	-0.0%	-0.1%	-0.0%	-0.1%	-0.0%

What the treatments do to the predictions

FD001, lifetime law fitted on the pooled fleet under each treatment

Carried through from the fitted parameters to the predictions anyone would act on, the ranking inverts. Counting the survivors as failures gives much the worst fit of the lifetime law, and the best remaining-life predictions of the three available treatments — better even than the oracle.

Two errors partially cancelling. The age-only estimator over-predicts remaining life when fed the correct law, and a law biased short pushes the predictions back down. The lesson is not that mishandling censoring is harmless. It is that downstream prediction error cannot decide whether the censoring was handled correctly; that question is settled on the parameters, against a planted truth.

treatment	beta	eta	MAE	bias	cover 90%
censored MLE	4.82	236.6	37.1	+21.4	91%
drop censored	4.41	225.0	32.5	+12.3	97%
as failures	2.93	188.8	27.9	-9.2	98%
oracle	4.55	224.8	32.5	+12.0	97%

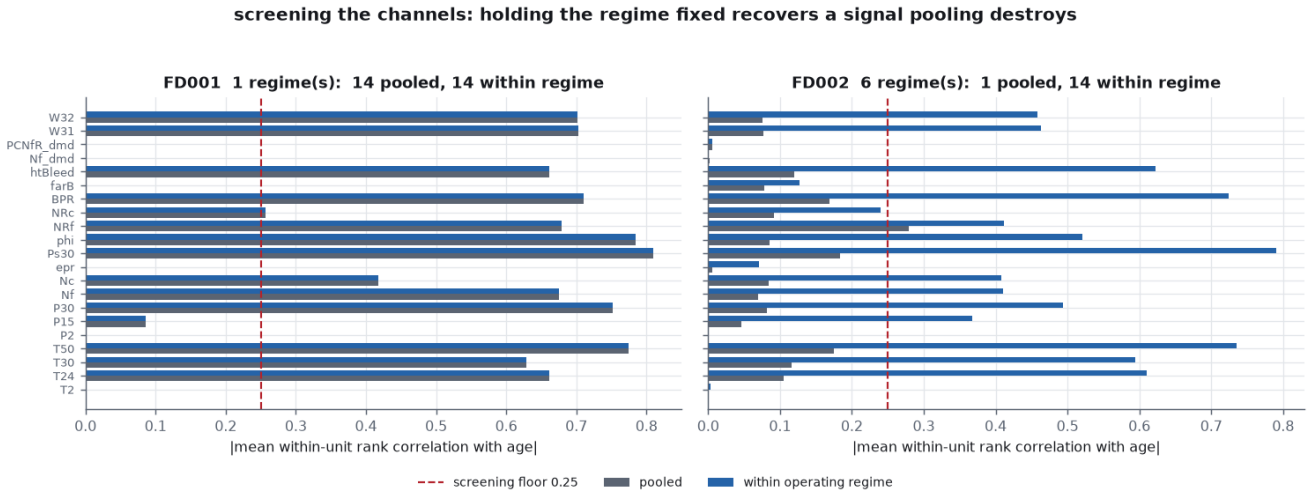
The oracle row uses the censored units' true failure cycles. It is the unobtainable target, not a method. Every row predicts the same engines from a law fitted to a fleet containing them, so the differences between rows are attributable to the treatment alone.

Which sensors carry signal, and where

mean within-unit rank correlation between channel and age

Screening the twenty-one channels by how strongly each moves with age gives two different answers depending on whether the operating regime is held fixed, and the difference is a result rather than a detail. On the single-regime subsets the two screenings agree exactly. On the six-regime subsets, pooled screening finds one usable channel out of twenty-one — no correlation in FD002 exceeds 0.28 — and the fleet looks as though it has almost no degradation signal in it. Holding the regime fixed recovers fourteen.

Moving the engine between six flight conditions swings every sensor by far more than degradation does, so pooling buries a real monotone drift under a much larger between-regime variance. Ps30 reads +0.18 pooled in FD002 and +0.79 once the regime is fixed.



subset	regimes	informative pooled	informative in regime	masked	max pooled |rho|	max regime |rho|
FD001	1	14 of 21	14 of 21	0	0.81	0.81
FD002	6	1 of 21	14 of 21	13	0.28	0.79
FD003	1	10 of 21	10 of 21	0	0.78	0.78
FD004	6	1 of 21	9 of 21	8	0.42	0.78

A second finding came from a channel that refused to stay excluded. The engine pressure ratio is exactly flat in FD001 and FD002 and trends at +0.31 and +0.21 in FD003 and FD004 — the two subsets that degrade the fan as well as the compressor. So channel usefulness depends on which fault is growing, not only on how the engine is flown, and a sensor set chosen on FD001 would discard the channel that marks the second failure mode.

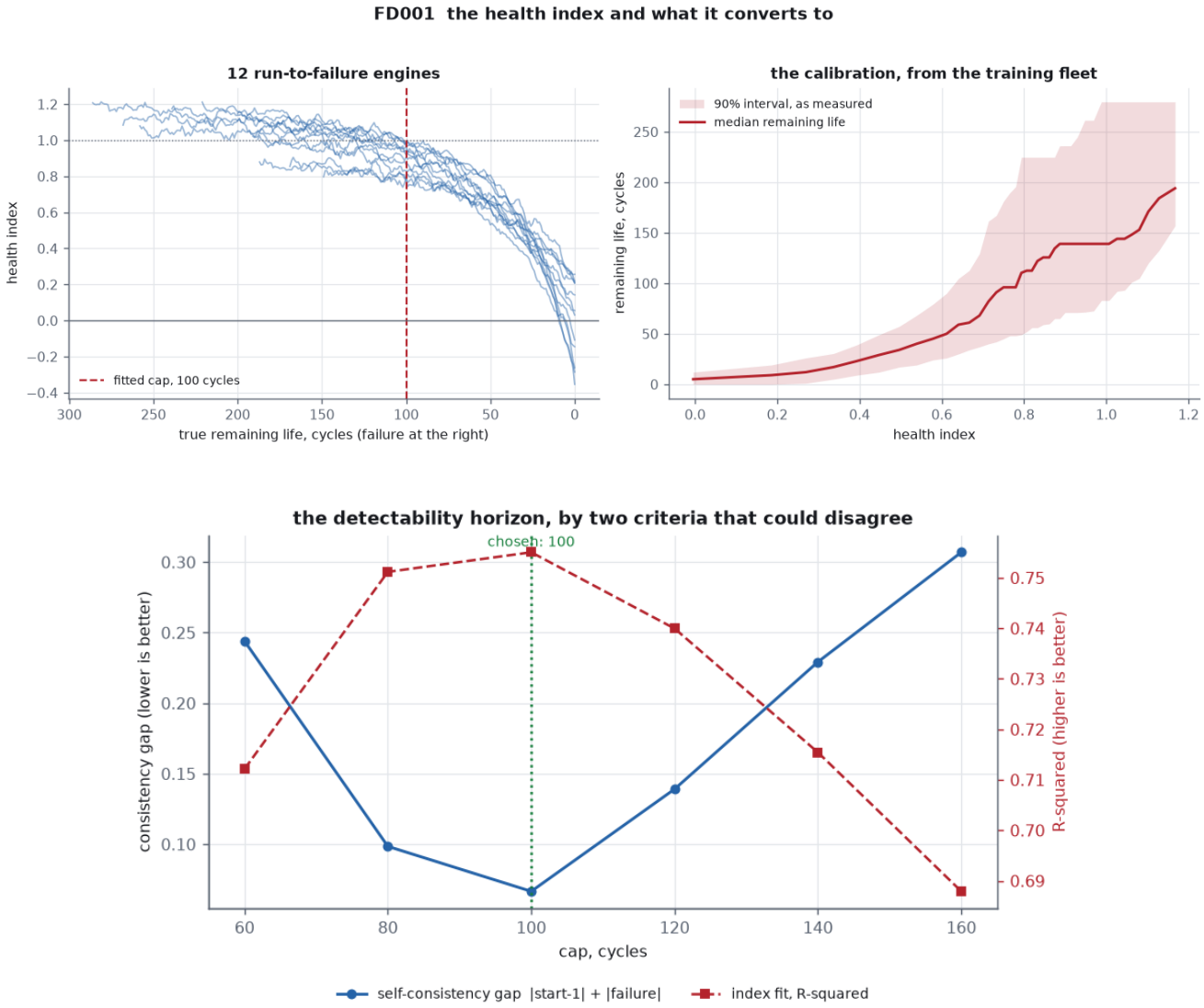
Estimator B: the degradation signal

FD001, cap 100 cycles, 14 channels, index R2 0.755

The second estimator is never told how long an engine lives — no lifetime law, no survival curve, not even the age. The channels are reduced to one health index by least squares against a target that sits at 1 while the engine is healthy and falls to 0 at failure, the index is smoothed causally, and it is converted to cycles through the empirical conditional distribution of true remaining life given health. The interval is that measured spread, not a variance formula.

The target needs a cap, because degradation has an onset: an engine does not begin dying at cycle 1, and a target declining from the first cycle asks the regression to find a trend that is not there. The cap is estimated rather than taken from the literature's usual 125, and two criteria that could disagree do not — the cap at which the index lands on 1 healthy and 0 dead without being asked to, and the cap maximising fit quality, agree on 100 cycles here.

Read physically, that cap is the horizon over which these instruments say anything at all. Beyond roughly a hundred cycles from failure they do not distinguish an engine with 150 cycles left from one with 250, which sets a ceiling on what any sensor-driven estimator can achieve.

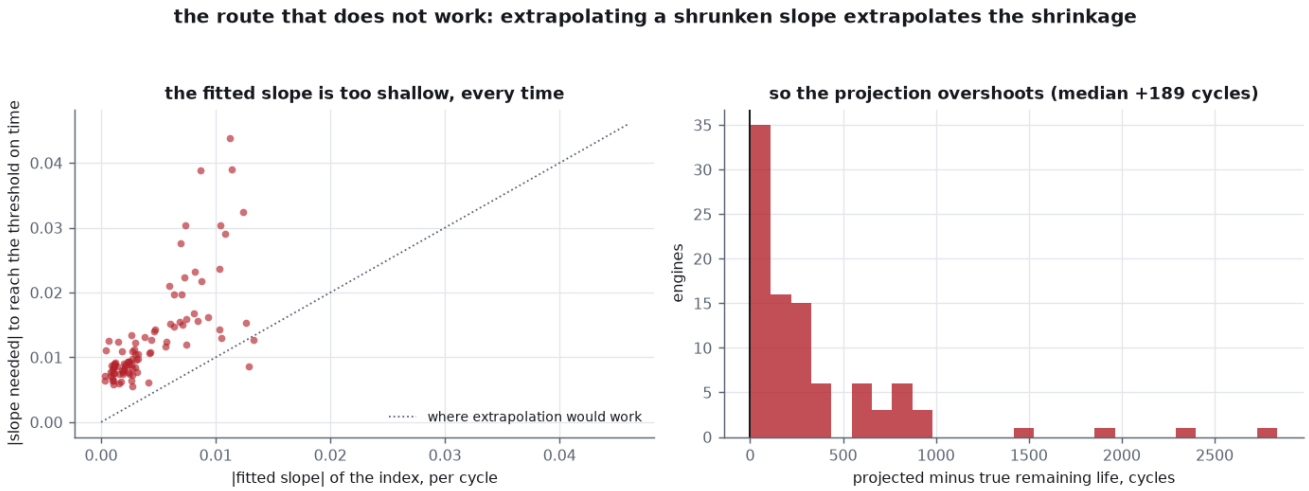


The route that does not work

FD001 test fleet, slope fitted on a 40-cycle trailing window

The obvious thing to do with a declining health index is to fit its slope and project it to the failure threshold. The fitted slope comes out too shallow by a factor of 3.6 at the median and by more than five on a quarter of the fleet, and not because of noise: the index carries a per-cycle noise of about 0.042 against an eleven-cycle mean, which is small. To reach the threshold on time the index would have to fall at about -0.009 per cycle, which is roughly what the run-to-failure engines average over their final hundred cycles — but the slope fitted on a test unit's trailing window has a median of -0.003, with half the fleet between -0.007 and -0.001.

The cause is attenuation. The index is a regression estimate of health, so it is shrunk toward the fleet mean, and a shrunken signal has a shallower slope than the quantity it estimates. Extrapolating a shrunken slope extrapolates the shrinkage. Degradation also accelerates, which compounds it: an engine censored while still at index 0.8 is in the slow early part of its decline, and the fall still to come is steeper than anything in its own history. Calibrating the inversion fixes both at once, because the calibration is fitted on the same shrunken index it is applied to.



A second bias was found and fixed on the way. The calibration was being fitted on a cycle-weighted population and applied to a unit-weighted one: an engine that lives 350 cycles contributes more than twice as many rows as one that lives 150, almost all from its long healthy stretch. At health 0.8 to 0.9 the median remaining life among training cycles is 130 against 95 among test units at the same health. Weighting each engine equally removes most of it; the remainder is cap saturation and is reported rather than corrected.

The two estimators against each other

FD001, mean absolute error in cycles

Neither estimator dominates, which is the point of having two. Close to failure the degradation signal is 4.4 times more accurate — 9 cycles against 41 — and that is the band with consequences attached. Its advantage is still about two to one out to fifty cycles, and then it collapses.

Beyond fifty cycles the two sit within a factor of two of each other and which one wins is not predictable from the horizon alone: the age law takes one or two bands on every one of the four fleets, but never the same ones. That is what the detectability horizon looks like from the outside. Past it the sensors have nothing left to add, the degradation estimator falls back on the spread of its top calibration bin, and the fleet's lifetime distribution is as good a guide as anything.

They disagree by 30 cycles on average and their predictions correlate at only +0.68, so they are not two tunings of one model. Where they disagree, which is right is predictable from the horizon rather than from which looks better.



true RUL	n	A, age only	B, degradation	closer
0-25	19	41.0	9.3	degradation
25-50	11	29.0	11.5	degradation
50-75	13	20.2	25.2	age law
75-100	23	31.8	30.8	degradation
100-125	23	37.5	21.0	degradation
125-150	11	26.8	12.6	degradation
all	100	32.5	19.6	degradation

Coverage, as measured

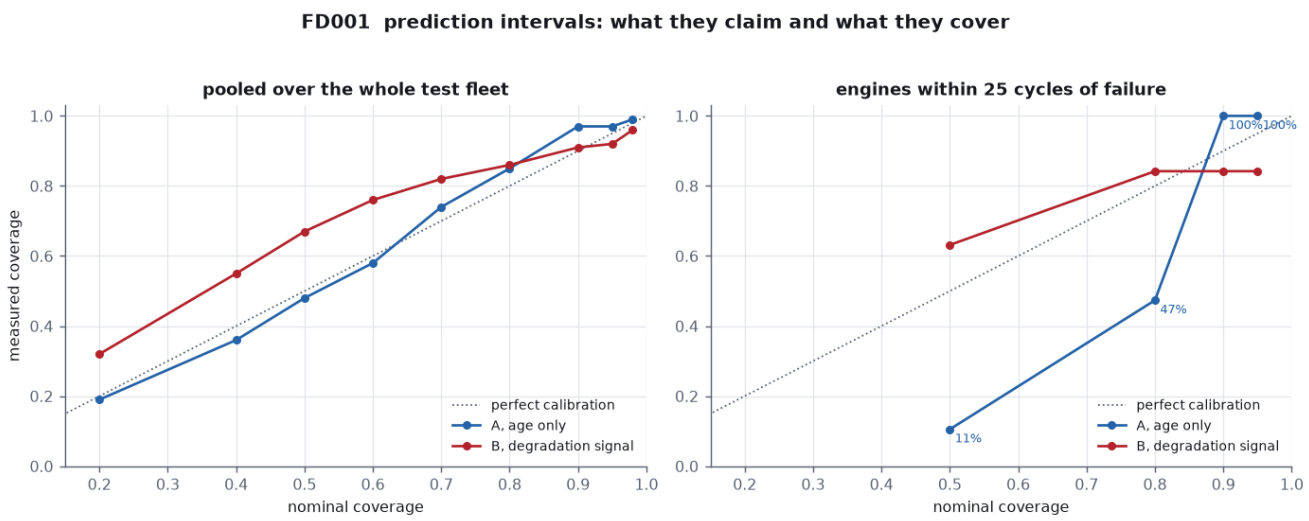
FD001, degradation signal; nominal levels fixed before any coverage was computed

A prediction interval is a claim with a number attached, and whether it holds is an empirical question. Pooled over the test fleet both estimators look broadly calibrated: the age law's nominal 90% interval covers 97% here and the degradation signal's covers 91%.

Conditioning on the horizon takes that picture apart, and it does so in two different ways on different fleets. On FD001's engines within 25 cycles of failure the age law's nominal 50% interval covers 11% and its nominal 80% covers 47%, while its nominal 90% covers 100% — the prediction is so far off that only the widest interval catches the truth at all, and it catches everything. On FD004 the same nominal 90% interval covers 53%.

Neither is calibration. The intervals are the wrong shape rather than uniformly too wide or too narrow, and pooling averages an interval that misses almost everything near failure against one that contains almost everything elsewhere, which comes out looking respectable.

> These are measured coverages, not intended ones. Where a nominal 90% interval covers 53% of a band, 53% is what appears. The degradation estimator's intervals hold up better because they are built from the conditional spread actually observed in the training fleet rather than from an assumed error law — which is a reason to prefer that construction, not evidence that the model is right.



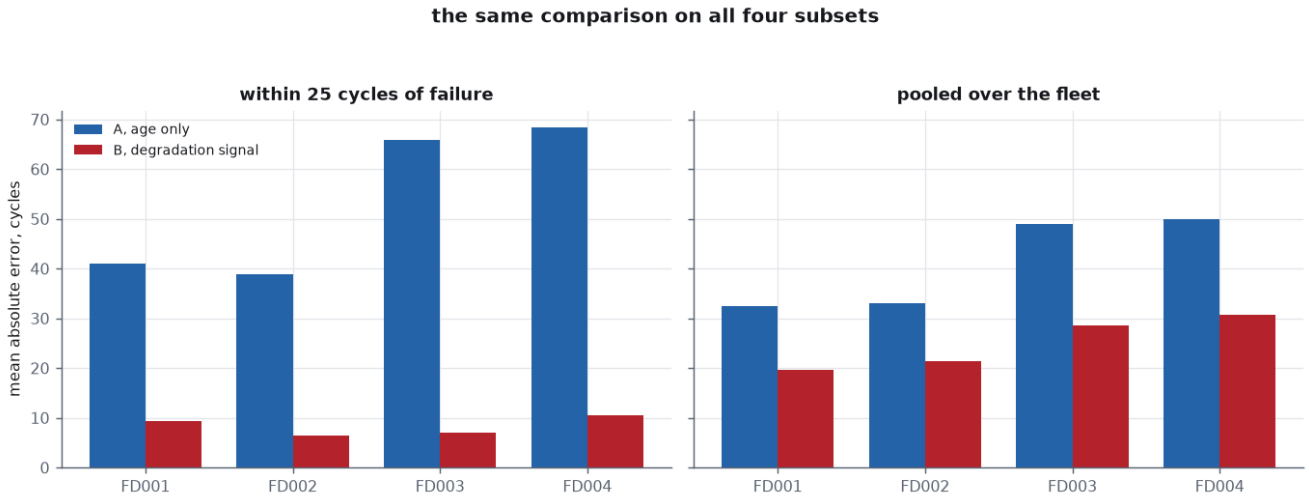
Coverage, as measured (continued)

true RUL	n	MAE	bias	optimistic	cover 50%	cover 90%
0-25	19	9.3	+7.2	74%	63%	84%
25-50	11	11.5	+7.0	64%	64%	91%
50-75	13	25.2	+18.2	62%	62%	85%
75-100	23	30.8	+25.9	91%	57%	96%
100-125	23	21.0	+12.2	65%	70%	91%
125-150	11	12.6	-2.9	64%	100%	100%
all	100	19.6	+13.0	72%	67%	91%

Does it hold across the four fleets

mean absolute error in cycles; coverage at nominal 90%

The subsets differ in operating conditions and fault modes, and the comparison holds on all four. The degradation signal is between four and nine times more accurate than the age law on the engines closest to failure, and wins pooled on every subset. The age law's intervals come apart near failure on all four, though not at the same nominal level: at 90% the shortfall shows on three of them and worsens with the fleet's difficulty, down to 53% on FD004 — six operating regimes and two fault modes, where the health index is also weakest and the fitted cap longest. On FD001 the nominal 90% interval instead covers 100% near failure and the failure appears at the narrower levels, where nominal 50% covers 11% against 48% pooled.



subset	A near failure	B near failure	A pooled	B pooled	A cover 90% near	A cover 90% pooled
FD001	41.0	9.3	32.5	19.6	100%	97%
FD002	38.8	6.4	33.1	21.5	84%	93%
FD003	65.8	7.0	48.9	28.6	73%	96%
FD004	68.3	10.6	49.9	30.7	53%	86%

Recovering a planted truth, and where it stops

90 run-to-failure and 90 censored simulated engines per row

Simulated fleets with a known health law, a known detectability horizon and known useless channels check that the estimator finds what is actually there. With a clean signal it recovers the planted horizon exactly and predicts remaining life inside it to within three cycles. As sensor noise rises the error grows, the recovered horizon drifts by a grid step, and the intervals widen to keep their coverage — which is the behaviour wanted, since an estimator losing its signal should become less certain rather than merely less accurate.

sensor noise	cap found	index R2	MAE inside horizon	cover 90%
0.10	100	0.995	2.9	96%
0.25	100	0.969	5.4	96%
0.60	120	0.845	9.2	94%
1.50	120	0.459	16.0	96%
3.00	120	0.169	26.8	94%

The last row is the useful one: at noise 3.0 the index explains little and the error is 27 cycles, so the method has stopped working — and says so through its intervals rather than through a confident wrong number.

What this does and does not establish

limitations

Established. Remaining-life error depends so strongly on the horizon that a pooled figure misdescribes both ends of a fleet, and pooled coverage can look calibrated while the intervals are badly overconfident exactly where decisions are made. A sensor-driven estimator is several times more accurate than an age-only one close to failure, and within a factor of two of it beyond fifty cycles, where which one wins is not predictable from the horizon. Which sensors carry signal depends on the operating regime and on the fault mode, so a channel selection is not transferable between fleets of the same engine.

Not established. These are simulated engines. C-MAPSS models degradation with a smooth growing fault and Gaussian sensor noise, and real fleets have maintenance actions, replaced parts, sensor drift and faults that arrive without warning, none of which is here. The detectability horizon of about a hundred cycles is a property of this simulator's instrumentation and should not be carried over to real hardware.

The censoring conclusion also deserves care. Dropping the censored units performs well here because of how the archive was built, and badly under the censoring real fleets have. The transferable result is the method for telling the difference — compare treatments against a planted truth under the censoring mechanism you actually face — not the ranking that method produced on this dataset.